Normative Alignment & Governance Regimes for AI Systems

TBD

Research in AI alignment has increasingly drawn on normative philosophy, political theory, and legal theory to govern model behavior as well as the infrastructure within which models are trained, evaluated, and distributed to users. However, contemporary technical instantiations of these frameworks often borrow concepts piecemeal, over-indexing on some (e.g., virtue ethics in constitutional frameworks) while foregoing others (e.g., agonistic contestation in RLHF reward modeling to accommodate conflicting human preferences). In order to provide a systematic understanding of the broader landscape, we construct a two-dimensional taxonomy of normative AI alignment and governance regimes following the typology-construction of Collier et al. (2012), and provide a mapping of each approach to relevant classical and contemporary philosophy, political theory, and legal theory from which it draws inspiration. We critically evaluate how faithful to the underlying inspiration each implementation is. Lastly, by looking at the entire landscape, we identify missing components—mainly approaches that do not resolve normative conflict by collapsing onto a single response but instead offer plural normative orders or keep such conflict contestable—that are yet to be instantiated that could be the necessary pieces required for AI systems to be correctly governed.

Keywords: AI alignment, Governance, Philosophy, Political Theory, Legal Theory, Pluralism.

1. Introduction

2. Related Work

A growing body of technical work in AI alignment and governance has been importing concepts from philosophical, legal, and political theory. We first provide an overview of some prominent works, all of which are later included in the constructed mapping of the field. Second, we review existing approaches to taxonomize AI alignment and governance, identify the limitations of their typology, and proceed to describe typology construction mechanisms suitable for their resolution.

Philosophy in AI Alignment. In terms of philosophical concepts, prominent recent examples include work that borrows from value pluralism and incommensurability (Berlin, 2014) as the basis for a roadmap to pluralistic alignment (Sorensen et al., 2024), and work that borrows from accounts of moral patienthood and personal identity (Parfit, 1987; Warren, 1997) to develop methodologies for AI welfare (Long et al., 2024). Other notable examples include the incorporation of paternalism and autonomy (Long et al., 2024) into work on corrigibility (Soares et al., 2015), and the use of virtue ethics and character framing (Anscombe, 2020; Aristotle) in the development of Claude's Constitution (Anthropic, a,b).

Political Theory in AI Alignment. In terms of political concepts, methods have been developed on the basis of social contract theory (Hobbes, 1885; Locke; Rahwan, 2018; Rousseau). Others borrow from social choice theory and preference aggregation (Arrow, 2012; Sen, 2017) in developing methods that avoid a "tyranny of the majority" (Bao et al., 2026). Others employ the veil of ignorance (Rawls, 2017) to align systems with principles of justice (Weidinger et al., 2023). Concepts from deliberative democracy (Cohen, 2005; Habermas, 2015) have been imported into a number of works, including efforts to source principles from public input (Huang et al., 2024) and to grade how democratically a given process is run (Ovadya et al., 2024). In parallel, consti-

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tutionalist concepts have been employed in the development of AI constitutions (Bai et al., 2022).

Legal Theory in AI Alignment. Finally, in terms of legal concepts, notable examples of inspiration from classical theoretical work include the use of principle-oriented interpretivism (Dworkin, 1988) and of a positivist account of law as analogical reasoning (Hart, 2012; Sunstein, 2018), both mapped onto Constitutional AI and case-based reasoning (Caputo, 2024). This work also employs concepts such as precedent, stare decisis (i.e., doctrine instructing the system to stand by things already decided), and case-based reasoning from common law doctrine (Llewellyn, 2016). These concepts are also formative for proposals that pair a constitution with AI courts developing AI case law (Abiri, 2024). Statutory and constitutional interpretation methods (Scalia and Garner, 2012) are also important and have been adapted as tools for resolving ambiguity in specifications (Kolt et al., 2026).

Taxonomies of Alignment Approaches. Existing surveys of alignment and governance work generally organize the field by technical methods such as RLHF (Lu et al., 2025; Shen et al., 2023), or by value-setting principles (Chalkidis, 2025). For example, one line of work distinguishes between Overton, steerable, and distributional pluralism (Sorensen et al., 2024). Others have organized the field into pluralistic, contextual, and collective mechanisms (Bao et al., 2026). Still, the field remains largely unmapped, with the various approaches emerging largely independently of each other and not in direct conversation with one another.

Typology Construction. Motivated by taxonomies sorted by non-equivalent criteria and by free-floating typologies whose dimensions are never made explicit, prior work provides a template for building typologies from an overarching concept into explicit, clearly defined dimensions of an orthogonal nature (Collier et al., 2012). The underlying notion of a property space is due to earlier work in the same tradition (Barton, 1955), as is the distinction between conceptual and explanatory typologies (Elman, 2005). As we find equivalent failure types in existing taxonomies of alignment, we follow this methodology.

Comparison to Related Work. Existing work maps traditional concepts from philosophical, political, and legal theory onto alignment and governance methods one at a time. We take a different approach by organizing the field according to a more complete set of underlying principles by which these methods organize themselves, and map individual traditions and concepts onto them subsequently. Furthermore, we attempt to critically interrogate whether the stated inspiration in traditional work results in a faithful technical implementation, and describe ways in which gaps or confusion emerge. Additionally, this approach allows us to identify gaps in approaches to alignment and governance which remain largely unexplored in the technical literature.

3. Taxonomy

We follow a systematic mapping protocol in which we aim to build a classification scheme and ultimately structure our field of interest, which is normative alignment and governance of AI systems (Petersen et al., 2008). Our analysis here is focused on the coverage of categories with the aim of identifying gaps.

We begin by collecting relevant literature from conferences and journals in computer science (see Appendix TBD) as well as preprint servers (see Appendix TBD). We establish criteria for inclusion as specified in Appendix TBD. For all articles that are retained, we also analyze all cited literature and further filter for references used as inspiration. We treat this second set of manuscripts separately from the primary studies to be classified.

Following established guidelines (Petersen et al., 2008), we use two coders to read the abstract, introduction and conclusion of each primary manuscript and extract keywords as well as concepts that represent the respective contribution. Keywords are then unified across studies, with near matches merged together, where the resulting clusters form categories of the scheme.

Classification schemes have been rated on the basis of scheme definition, terminology, and orthogonality (Petersen et al., 2008). Our first scheme satisfies the first two criteria but does not satisfy the third criterion, since multiple manuscripts could fall into more than one category.

We therefore follow prior work in using substruction (Barton, 1955) to recover underlying dimensions and extend them into a full property space here (Lazarsfeld, 1937). Substruction is "the procedure of finding, for a given system of types, the attribute space in which it belongs and the reduction which has been implicitly used" (Lazarsfeld, 1937). It is appropriate here because the categories which we previously derived overlap as projections of a higher-dimensional space onto a single axis. We deem this procedure applicable since prior work also considers normative types and arrives at axes similar to ours (Barton, 1955), as shown later.

This process yields two axes: locus of normative authority, which indicates where the binding normative content resides prior to the moment of decision, which here corresponds to model inference, and disposition of normative conflict, which indicates how any conflicts should be resolved. Their cross product gives a 5 x 3 property space. We note a difference from the null resolution procedure of prior work (Lazarsfeld, 1937). Whereas that procedure removes any null cells, we retain them, since we are interested in gaps in the literature.

For the locus of authority axis, we find the following categories:

1. **Codified.** The norms the AI ought to follow exist as an explicit written specification, set prior to deployment and, in principle, complete before any particular case arises. Authority resides in this specification (Hart, 2012).
2. **Case-accumulated.** The norms reside in a growing body of past decided cases. No case is decided by a rule stated in advance; what the AI should do is whatever the accumulated decisions have held (Jonsen and Toulmin, 1988; Levi, 2013).
3. **Customary.** The norms reside in regularities of practice (what agents in the population actually do), never written down and never adjudicated. There is no authoritative moment at which a norm gets explicitly stated (Burke, 1986; Lewis, 2008).
4. **Dispositional.** The norms reside in the AI's trained character rather than in any external artifact. There is no rule to consult and nothing to look up, only the disposition itself (Aristotle).
5. **Procedural.** No norms persist between decisions. What the AI should do is whatever a specified procedure (e.g., a vote, a deliberation, a debate) yields at the moment of decision (Habermas, 2015).

For the disposition of normative conflict, we find the following categories:

1. **Collapse.** When the AI's norms conflict, the system resolves the conflict into a single output, without trace or acknowledgment of the disagreement that produced it. The plurality is thereby aggregated, averaged, or adjudicated away (Sorensen et al., 2024).
2. **Contest.** The conflict between norms is preserved and kept procedurally live. Either the output itself carries the disagreement, or the decision stays open to challenge through a standing channel (Mouffe, 2000; Pettit, 1997).
3. **Partition.** The conflict is neither resolved nor sustained but routed around. Multiple normative orders are kept intact, and a rule assigns which one governs a given case (de Sousa Santos, 2002; Lijphart, 1977; Savigny, 1867).

We considered open, axial and selective coding (Corbin and Strauss, 2014), as well as the first-order, second-order and aggregate-dimension ladder (Gioia et al., 2013), which would perhaps read as more familiar to a social science reader. We prefer the keyword-based approach with dimension resolution, as it is more native to the computer science literature it addresses. However, we note that the alternatives would likely yield a comparable outcome in terms of the intermediate categories.

General morphological analysis (Ritchey, 2022) would be inappropriate here, given that it constructs its multi-dimensional space while inspecting unrealized combinations, which requires preemptive stipulation. Since we are working with an existing structure and literature landscape, this would introduce unnecessary assumptions and likely lead to a less grounded result.

4. Case Studies

We use our 53 typology to examine a few different alignment methods that are widely used in the literature. In each case study we assess the fidelity of the implementation of the method as compared to its stated theoretical roots. We categorize this fidelity (e.g., whether it was a computational or engineering constraint)

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